Mathematical Typesetting in LaTeX

Mathematical Expressions and Tables

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Mathematical Expressions

LaTeX provides environments to present mathematical content clearly.

Piecewise Function

$$f(x) = \begin{cases} x^2 + 2x + 1, & x < 0, \\ \sqrt{x+1}, & 0 \leq x < 4, \\ \dfrac{x^2 - 1}{x - 1}, & x \geq 4. \end{cases}$$

Matrix Representation

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Creating Tables in LaTeX

Tables can be formatted using alignment, borders, merged cells and colour.

Student	Midterm	Final
Amit	78	89
Neha	85	92
Rahul	91	94

- `tabular` creates rows and columns.
- `\multicolumn` and `\multirow` merge cells.
- `xcolor` adds colour to rows and cells.
- `booktabs` helps create clean, professional tables.

Conclusion

- Mathematical environments help present equations, matrices and functions systematically.
- Table environments organise data in a clear and readable format.
- Packages extend LaTeX with features such as colour, merged cells and professional table rules.

Thank You!